PROJECT DOCUMENTATION

Healthcare Data Analysis and Visualization Dashboard

1. Project Overview

This project focuses on analysing healthcare and patient data using Microsoft Excel and Power BI. The main objective of the project is to clean and prepare raw healthcare data, analyse patient information, and create an interactive dashboard to identify meaningful patterns and trends.

The analysis provides insights into patient demographics, medical conditions, admission types, hospital information, billing amounts, insurance providers, test results, and length of hospital stay. Interactive visualizations and slicers were used to make the analysis easier to understand and explore.

2. Tools Used

- Microsoft Excel
- Microsoft Power BI

3. Dataset

Source:

Healthcare patient dataset was obtained from a publicly available online source through Google search.

Data Contains:

The dataset contains information about patients and their hospital records, including:

• First Name	• Insurance Provider
• Last Name	• Billing Amount
• Age	• Room Number
• Gender	• Admission Type
• Blood Type	• Patient Type
• Medical Condition	• Discharge Date
• Date of Admission	• Length of Stay
• Doctor	• Medication
• Hospital	• Test Results

The dataset was used to analyse patient demographics, medical conditions, hospital admissions, billing information, and patient outcomes.

4. Steps Followed

The raw healthcare dataset was first cleaned and prepared in Microsoft Excel before importing it into Power BI.

The following data cleaning activities were performed:

1. Removed unnecessary or irrelevant data: Reviewed the dataset and removed any unnecessary information that was not required for the analysis.

2. Checked for blank and missing values: The dataset was checked for blank or missing values in important columns.

3. Corrected data formats: The data types of different columns were checked and corrected.

For example:

Age → Number

Billing Amount → Decimal Number

4. Standardised text data

Text fields such as:

Gender

Blood Type etc.

5. Checked duplicate records: The dataset was checked for duplicate or repeated records to ensure that the analysis was not affected by unnecessary duplicate entries.

6. Cleaned and formatted column names: Column names were reviewed and kept consistent and understandable for further analysis.

7. Created Age Groups: Age groups were created to make it easier to analyse patients based on age categories.

8. Prepared the cleaned dataset: After completing the cleaning and preprocessing steps, the cleaned dataset was saved and imported into Power BI for visualization and analysis.

9. Checked the data types and summarization settings in Power BI.

Created KPI cards to display important healthcare metrics.

Added slicers for interactive filtering.

Created charts and visualizations to analyse patient and hospital data.

Arranged the visualizations into an interactive healthcare dashboard.

Added titles and formatting to improve dashboard readability.

5. Key Insights

The healthcare dashboard provides several insights into the patient and hospital data.

The dashboard helps identify the most common medical conditions among the patients.

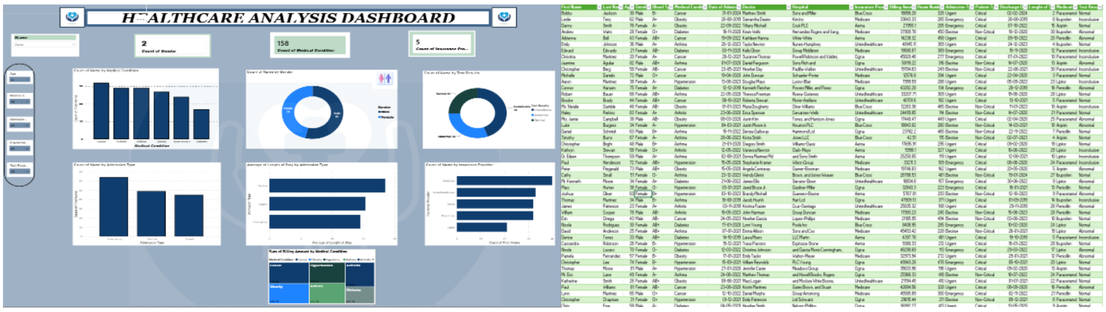
The distribution of patients across male and female genders can be analysed using the gender visualization.

Different age groups can be compared to identify the age range with the highest number of patients.

Emergency, Urgent, and Elective admissions can be compared to understand admission patterns.

The dashboard helps analyse the average billing amount associated with different medical conditions.

6. Screenshots



7. Files Included

https://docs.google.com/spreadsheets/d/14mG94wEXGmM_Rao3_UE-vsHwalMGqCMH/edit?usp=sharing&ouid=100728687848531959364&rtpof=true&sd=true – Cleaned healthcare dataset prepared in Excel.

https://drive.google.com/file/d/12a5pVmZ1rduknXvDFIodFuYXW7DnVoo1/view?usp=sharing – Power BI report containing the healthcare visualizations and interactive dashboard.

8. How to Use

https://docs.google.com/spreadsheets/d/14mG94wEXGmM_Rao3_UE-vsHwalMGqCMH/edit?usp=sharing&ouid=100728687848531959364&rtpof=true&sd=true to view the cleaned data.

https://drive.google.com/file/d/12a5pVmZ1rduknXvDFIodFuYXW7DnVoo1/view?usp=sharing in Power BI Desktop to explore the visuals

9. Conclusion

The Healthcare Data Analysis and Visualization project successfully transformed raw patient data into an interactive and informative Power BI dashboard. The project involved data cleaning and preprocessing in Excel, followed by data analysis and visualization in Power BI.

The dashboard provides a clear overview of patient demographics, medical conditions, admission patterns, hospital information, billing amounts, insurance providers, test results, and length of stay. The use of KPI cards, charts, and interactive slicers allows users to explore the data from different perspectives and identify important patterns.

Overall, this project demonstrates how Excel and Power BI can be used to clean, analyse, and visualize healthcare data effectively, helping users understand complex data and support data-driven decision-making.